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Role of Information and Communication Technologies in the Smart Grid

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ABSTRACT

In the smart grid operation and management, reliable and real-time information and communication networks play a very critical role. By integrating the appropriate information and communication technologies (ICT) infrastructure, automated control, sensing and metering technologies, and energy management techniques, the smart grid has emerged as a solution that empowers utilities and consumers to share the responsibilities of operating and managing the power grid more efficiently. This paper presents a survey on the role of ICT in the Smart Grid and Micro grid evolution from conceptual models to implementations.

Keywords: Smart grid, micro grid, information technology, wireless communications, renewable energy resources.

1. INTRODUCTION

The smart grid is a modern electrical power grid infrastructure for better efficiency, reliability, with possible integration of renewable and alternate energy sources. In order to achieve those broad objectives, smart grid integrates advanced information and communications technologies (ICT), automation, sensing and metering technologies, and energy management techniques based on the optimization of energy demand and supply into traditional power grid in order to make it more efficient in many ways. Smart grid offers better communications among all stakeholders in the system. Smart grid requires communications to be real-time in many cases. The ICT infrastructure in the smart grid environment needs to be reliable, highly-available, scalable, secure, and easy-to-manage. Power engineering society, along with its counterparts in ICT, has developed the first smart grid conceptual model which consists of three layers: energy and power systems layer, communications layer, and information technology layer [1-3]. This categorization is based upon from the seven-cloud conceptual model that was proposed by NIST [1], as shown in Figure 1. The ICT layers constitute about 70 % of the smart grid infrastructure [4]. From the software point of view, the conceptual model can be divided into three interconnecting and overlapping layers namely; Automatic Meter Reading (AMR), Advanced Metering Infrastructure (AMI) and Advanced Metering Infrastructure Plus (AMI⁺) layers as shown in Figure 2.

The smart grid ICT industries, such as IBM, Intel, Cisco, Oracle and Google, are all contributing substantially in the makeover process from traditional grid to smart grid utilities [5-9]. Moreover, academic institutions around the world are joining hands to model, build, implement and

operate the smart grid [10-15]. There has been a tremendous interest in the smart grid technology, as indicated by a jump from 3 Million to 45 Million hits of Google search results using the “smart grid technology” keyword over the last two years.

This paper presents a survey on the role of ICT in the smart grid evolution from conceptual models to implementations. It highlights the renewable energy integration in the smart grid, the viability of the various communications technologies, and the migration path from smart grid to micro grid.

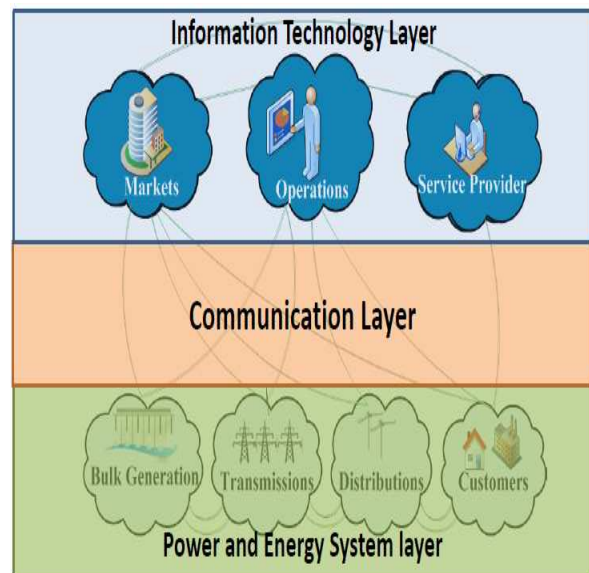


Figure 1: Smart Grid Conceptual Model



Figure 2: Smart Grid Software Layers

2. RENEWABLE ENERGY INTEGRATION TO THE SMART GRID

Traditional power generation plants are centralized and mostly are at far distances from the consumption premises. General speaking, most generated power is coming from non-renewable, non-variable resources such as oil, gas, coal and nuclear energy [1, 5]. Within the smart grid context, renewable non-variable resources such as pump storage, geothermal, biomass and hydro are used more than before. More generation from renewable variable sources, such as wind and solar energy as well as Plug-in-Hyper-Vehicle (PIHV), are added at the centralized generational stations as well as at the consumption cites, as shown in Figure 3.

According to the U.S. Energy Information Administration (EIA) International Energy outlook 2011 report, the renewable energy resources will be used to generate more power [16]. The report showed that North American and Europe are inviting in renewable to generate power where it is needed. The report forecasted that renewable energy will surplus all other energy resources in Europe by the year 2035 [16]. One of the major factors will play a role in the renewable energy integration is the ICT utilizing.

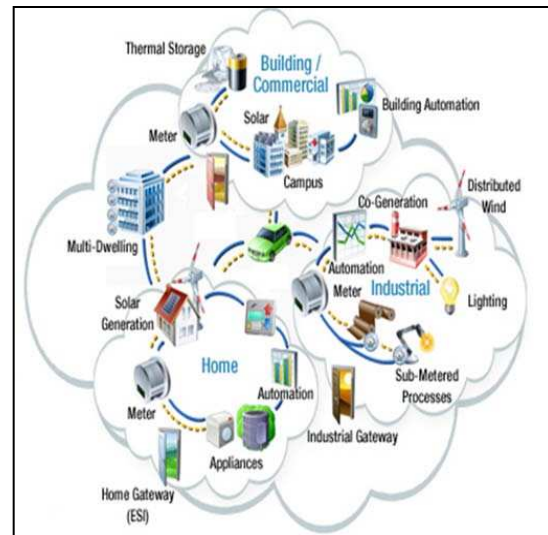
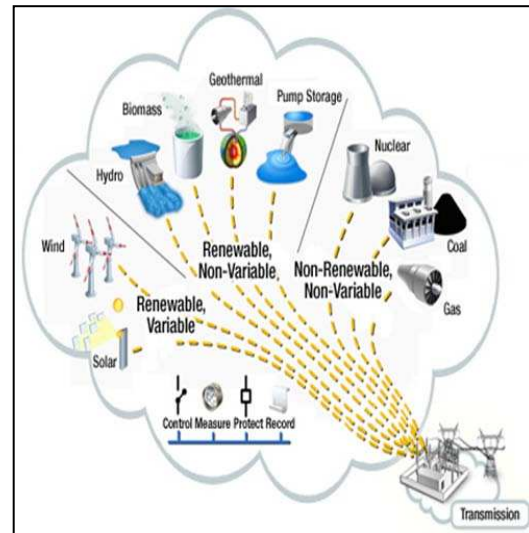


Figure 3: Smart grid centralized and distributed generations [1].

3. ROLE OF INFORMATION TECHNOLOGY IN SMART GRID

The role of information technology is one of the major factors that are transforming the traditional grid into smart grid. The smart grid information technology layer consists of computing platforms, operational systems, business applications and business services. Energy management, transmission operations, distribution operations, independent system operator (ISO), regional transmission organization operations (RTO) are among the major functions that constitutes the role of the IT layer [4-7].

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a. Computing Platform and Operational Systems Layer

This layer consists of set of high end servers that host all grid operation systems such as Real-time data from sensors outside

- Grid optimization, switching plans, outage information, geographic information.
- Tagging power flow analysis, dispatching and tracking.
- Demand side management and demand response.
- Volt-Var predictive and faulty locations, protection and real-time SCADA analysis.
- Self-healing and correction of disturbances on the grid.
- Transmission and distribution automation, communication networks analysis and management.
- Cyber Security protection, renewable energy integration process.

b. Business Applications and Services Layer

This layer consists of set of software packages that handle various business applications and services such:

- Utilities customer care and billing, consumer interface and web interface.
- Business and home energy management, distribution mobile workforce management
- Third party service providers, third party access for marketing and financial applications.

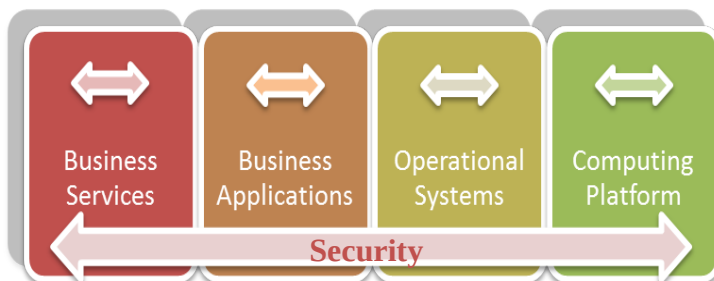


Figure 4: Show Smart Grid information technology layer building blocks.

4. ROLE OF COMMUNICATION NETWORKS IN THE SMART GRID

The communication networks are the nerve and vines system in the whole smart grid infrastructure. They connect power system generations, transmissions, distributions and consumptions systems to the information and operation software platforms that were highlighted in section 3. Information containing the real-time data that

reflects the health and status of the entire grid is transmitted and exchanged bi-directionally among the grid operators, consumers and ecosystem operators.

The smart grid communication networks can be divided into three sub-layers: AMR networks layer, AMI networks layer and AMI+ networks layer in addition to the existing SCADA. Each network serves set of devices throughout the whole grid [11-14, 17-33]. Each network can be classified based on service orientation, device type that is connected and the communication media. The smart grid communications networks can be classified as the followings [4-5, 24- 25].

a. Consumer Premises Networks (CPNs)

These networks are part of the AMR networks. They provide the communication between the appliances and equipment with the consumers' premises. Based on the consumer's consumption profile, these networks can be divided into three different networks namely; Home Area Networks (HAN), Business Area Networks (BAN) and Industrial Area Networks (IAN) [4-7, 12, 25-27].

The devices that are served by these networks can be:

- Smart home appliances, In-home display, Home computing devices.
- Load control devices, Demand Side Management, Power Measurements
- Electrical vehicles outlets, Renewable energy integration

The communication technologies that are utilized in these networks are wired and wireless networks. Examples of such networks are Zigbee, Xbee, Bluetooth, Z-Wave, Wi-Fi, Home Plug, 6-lowPAN, BACnet and SAEJ2847 [4-7, 28 -30].

The smart energy meter data exchange through these networks can be real-time measurement parameters (RTMP) or power consumption data. The RTMP are voltage, current, power and frequencies and they are used to check continuity and quality of the power. These measurements can be utilized in demand side management and demined response. Table 1 summarizes the consumption data profile for a smart meter [28].

Table 1: Smart Meter Traffic Analysis- IEEE Std 2030-2011 [28]

Description	Connects smart meter through wireless NAN. Smart meters could be residential (including building/business) or industrial-grade.
Latency	Highly variable latency requirements and sensitivities, depending on application. End-to-end application information transfer time: < 4 ms to 15 s
Bandwidth	50 kHz to 40 MHz (but others are possible)
Bitrate	1 Kbps to 30 Mbps
Payload	Size: 10 bytes to 1500 bytes, Frequency: 50 000 packets per second to 1 packet per minute
Quality of service	protocols mentioned above contain some type of quality of service/differentiated services mechanism
Reliability	Reliability Non-essential, important, or essential—depending on application.
Data occurrence interval	Data occurrence interval Could be periodic or aperiodic, depending on the application. This interval could range from 1 ms to 30 min.

b. Neighborhood Area Network (NAN)

This network is part of the AMR and AMI network. It collects the information from the consumers' devices through CPNs via the smart meter and passes it to the utility data centre for further processing and feedback actions. The devices that are served by these networks can be utilized for [4-5]:

- Advanced smart meters such energy consumption and RTMP, load control relays.
- Concentrators (collection point for meters from all neighborhoods).

The communication technologies that are utilized in these networks are wired and wireless networks. Examples of such networks are RF Mesh, RF Radio point-to-multiple point, Wi-Fi, WiMax, LTE, GPRS/EDGE, PLC/BPLC, and FTTP/FTTH/Ethernet [5, 28- 30]:

c. Access Area Network

This network is responsible for grid connected at the distribution level. The devices that are served by this network can be [5]:

- Voltage Regulators, re-closers and remotely operable switches.

- Capacitors banks and active power factor connection, distance to fault relays and line fault indicators.
- Phasor measurement and distributed RTUs, line sag indicators and maximum demand indicators.
- Renewable energy resources

The communication technologies that are utilized in these networks are wired and wireless networks; examples of such networks are WiMAX, LTE, GSM-CDMA, 3G/LTE Public, BPLC and FTTP/FTTH/Ethernet [17-18, 28- 31]:

d. Backhaul Network

This network is responsible for grid connected at the distribution level. The devices that are served by this network can be [5]:

- SCADA –RTUs substations, Intelligent Electronics Devices (IDE)
- Protection Relays, Oil level, Pressure and Temperature Sensors, Monitoring Cameras

The communication technologies that are utilized in this network are wired and wireless networks. Examples of such networks are LTE/ LTE Public, WiMAX, Microwave, Fiber, BPLC and FTTP/FTTH/Ethernet [28-32].

e. Core and Office Network

This network is responsible the corporate communications for services such as:

- Corporate communications-Voice and Data.
- Corporate communication-Planning and QoS.

The communication technologies that are utilized in this network are wired and wireless networks. Examples of such networks are Leased Line Circuits, GPRS and LTE, Fiber and FTTP/FTTH/Ethernet [27-32].

f. External Access Networks.

These networks are the public access networks that give access to the ecosystems providers' access to some of the above networks. The external networks mostly utilize public access networks. Figure 5 shows the networks options available for the smart grid. Table 2 shows the wireless communication networks key parameters and related smart grid applications [4, 12, 17, 20].

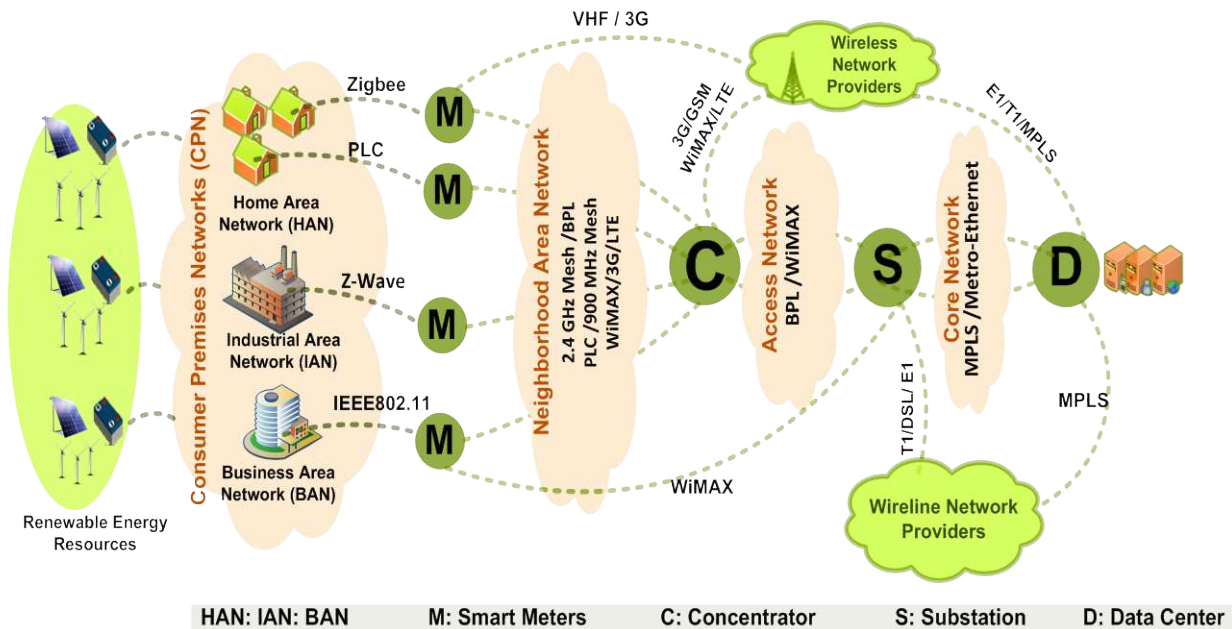


Table 2: shows the smart grid communication networks parameters and role

Wireless Technology	Data Rate	Approx. Coverage	Applications
Bluetooth	721Kbps	1-100m	Meters (AMI), HAN, BAN, IAN
IEEE 802.11 (Wi-Fi)	1-54Mbps	100m	Meters (AMI), Distribution Automation [DA]
Z-Wave	40 -250 kbps	30 meters point-to-point, unlimited with mesh	Meters (AMI), HAN, BAN, IAN

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		networking	
IEEE 802.15.4 (ZigBee/6LoWPAN)	250 Kbps	100+ meters	Meters (AMI), HAN, BAN, IAN
GSM	Up to 14.4Kpbs	1-10km	AMI, Demand Response, HAN, BAN, IAN
GPRS	Up-to 170kps	1-10km	AMI, Demand Response, HAN, BAN, IAN
WiMAX	DL: 46 Mbps UL: 4 Mbps At 10 MHz TDD	Up to 50 km	Meters(AMI), DA, Mobile workforce management
LTE	DL: 302 Mbps UL: 75 Mbps at 20 MHz FDD	5 – 100 km with slight degradation after 30km	Meters (AMI), DA, Mobile workforce management
MobileFi	20Mbps	Vehicular Std.	AMR, HAN, BAN, IAN

5. MIGRATION FROM SMART GRID TO MICROGRID

As the smart grid is shifting from conceptual modeling phase to real implementation, scalability, reliability, and cyber security among others factors are becoming major issues. The need for more effective and efficient smaller smart grid that has decentralized generation and storage facilities located physically near the consumption premises is emerging nowadays.

There are several definitions of the term “micro grid” cited in the literature. However, in this work we are interested in a small self-contained power system, operating at the consumer voltage, in a single geographical location, where generators and loads are grouped together, where some renewable generation is used, and where local control over these resources is exerted at a local level [33].

Generally speaking, the smart micro grid concept is the division of a large scale power grid into small community power grids. Each of these small community grid contains local traditional power generation and renewable energy resources; i.e. dividing the large metropolitan smart grid area into smaller segments. Each of the segments is an autonomous stand-alone grid that can be called micro grid. This micro grid can be managed locally and

can be integrated to the larger grid to trade power and exchange information when needed. Micro grid benefits are many and few to mention are:

- Power is generated where it is needed.
- Ease of renewable energy integration
- Local control and ownership
- Reduction in transmission losses.
- Reduction in Carbon footprint.
- Stand-alone grid and operate in isolation mode in case of blackout.
- Easy to finance and faster to build.

There are many successful smart micro grid models and implementations. Figure 6 reports an Australian micro grid model [39] and Figure 7 detailed a Canadian model [21] and many others to mention [35-38]. Closer look to both models, one can identify the load mix power generations, local storage, local controller and the gateway to integrate the micro grid to the larger utility grid.

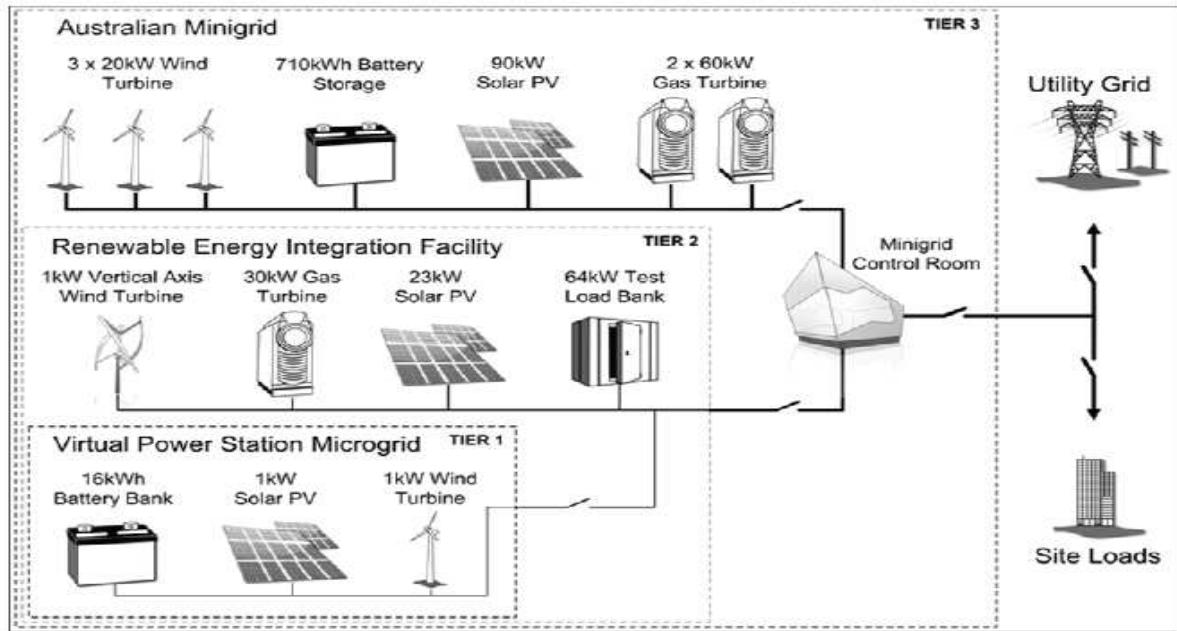


Figure 6: Australian smart micro grid model [39]

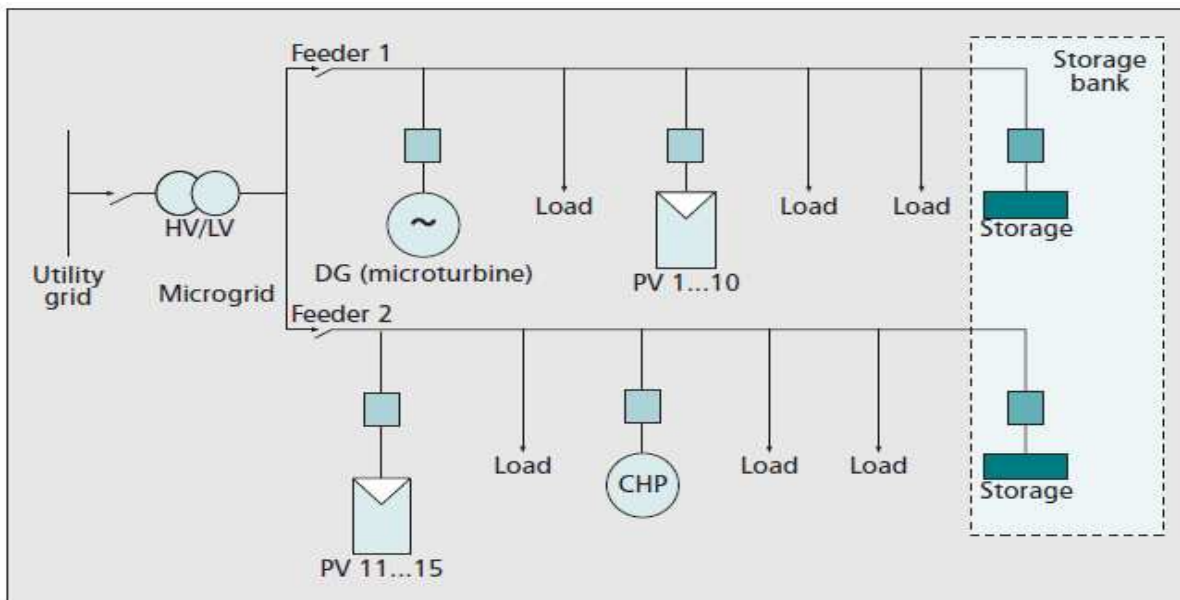


Figure 7: Canadian smart micro grid model [21].

6. REMARKS AND CONCLUSIONS

This paper describes the smart grid evolution from conceptual model to implementation. The role of the smart grid enabling technologies namely, information and telecommunications, is highlighted.

Power generations, transmissions, distributions and consumptions enabled with the ICT empower the shareholders for better communicate in two directions and manage the grid efficiently.

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The need for smart micro grid is explored along with justifications and benefits of the transformation from smart grid into smart micro grids. Models of the smart grid and smart micro grid are presented to appreciate the underlying technologies.

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